Li-Yu Chu

Vancouver, BC • liyu.chu.work@gmail.com
Available from mid-December 2026 • Open work permit (PGWP) valid to December 2029
https://www.linkedin.com/in/chuliyu/ • https://luichu.dev/

Software Developer | Java / Spring Boot Backend

Summary

Backend developer with three years of professional Java/Spring Boot experience building internet banking features at HiTrust, focused on transaction recovery, retry handling and consistent payment state. Builds personal projects that apply the same controls to AI tools and distributed systems, in TypeScript, Go and Python. Earned **AWS Certified Solutions Architect – Associate** and **HashiCorp Certified: Terraform Associate (003)**.

Experience

HiTrust, Inc. Taipei, Taiwan
Software Engineer January 2022 – December 2024

- Developed **Java/Spring Boot** internet banking features and REST APIs for Angular/TypeScript front ends.
- Built recovery and synchronization logic so failed or retried banking requests kept transactions consistent, with response-driven status updates and failure-state transitions.
- Developed foreign-currency SQL and logic for scheduled payment/debit and retry workflows, including core-banking debit requests; translated banking rules and message formats into application behaviour.
- Improved SQL and **Hibernate/JPA** data access on Oracle and SQL Server, including migration from JDBC.
- Wrote **unit and integration tests** (JUnit), took part in code reviews, redrew workflow diagrams from traced code paths, and resolved integration issues with QA and bank acceptance-testing teams.

Astra Technology Taipei, Taiwan
Product Planner (Data) March 2019 – December 2021

- Used Python/pandas and SQL for time-series forecasting; turned findings into product requirements and specifications.

Projects

Google Tasks MCP AI tool integration (public source)
TypeScript, Node.js, MCP, Zod, Google OAuth/REST, Vitest • Repository

- Built an MCP server giving AI clients authenticated Google Tasks access, with Zod input schemas and paginated retrieval; the live MCP surface is read-only.
- Tested approval-gated writes, account isolation and duplicate-action prevention with synthetic integration tests.

Raft-Recovery Educational distributed queue
Go, gRPC, Raft, write-ahead log, snapshots • Repository

- Implemented a durable queue with checksummed write-ahead logging, snapshots and replay, then a three-node replicated version with persistent Raft state, majority commit and snapshot catch-up.
- Added worker leases, heartbeats and attempt fencing; designed for at-least-once delivery, validated in local scenarios.

Arc Sentinel Wallet activity indexing
Python, FastAPI, PostgreSQL, SQLAlchemy, Docker • Repository

- Built wallet/USDC indexing with a confirmation buffer, retry/backoff and RPC fallback; uniqueness constraints, checkpoint transactions and advisory locking keep re-runs and competing indexers from duplicating records.

Technical Skills

Backend: Java, Spring Boot, Hibernate/JPA, REST APIs, SQL, Python, FastAPI, Go, TypeScript, Node.js
Data: Oracle, Microsoft SQL Server, PostgreSQL, Redis, Airflow, pandas
Cloud & delivery: AWS Lambda, API Gateway, DynamoDB (Chainy), Terraform, Docker, GitHub Actions
Testing & AI tools: JUnit, pytest, Vitest; Codex, Claude Code, ChatGPT, GitHub Copilot, Cursor, Ollama

Education & Certifications

Fairleigh Dickinson University Vancouver, BC
M.S. in Applied Computer Science Expected December 2026

National Taiwan Normal University Taiwan
B.A. in Geography 2011 – 2015

AWS Certified Solutions Architect – Associate • HashiCorp Certified: Terraform Associate (003)